SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning,
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 4 **Duration: 1 Hour**
Total: Marks:40

Annexure A

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Code of Conduct:

I __T. Niranjan Kumar (192473007)__ (Name / Reg No) certify that this submission is my original work and that I have adhered
to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

***Signature of the student:** __T. Niranjan__*

INDUSTRY SCENARIO

A healthcare company has collected patient records containing age, blood pressure, cholesterol, glucose level, BMI and family history. The objective is to build a predictive model for early diabetes diagnosis and an AI agent that recommends personalised treatment actions such as Diet, Exercise, Medication and Monitor based on patient state transitions over time.

TASK 1 – DATA PREPARATION & INDUCTIVE LEARNING

(a) Inductive Learning Approach

Inductive learning is suitable because the model learns a general rule from many patient examples. Historical patient records are used as training examples, and the learned pattern is then used to predict diabetes for a new patient.

Target Variable

- Diabetes Outcome / Diagnosis: Diabetic or Non-Diabetic.

Key Features with Justification

- Glucose Level: Directly reflects blood sugar and is an important indicator of diabetes risk.
- BMI: Higher BMI can be associated with increased diabetes risk.
- Age: Risk can vary with age and therefore contributes useful predictive information.
- Family History: A family history of diabetes can indicate hereditary risk.

Blood pressure and cholesterol can also be included because they provide additional health information for the prediction model.

(b) Handling Class Imbalance

If the dataset contains more non-diabetic than diabetic cases, a model may become biased toward predicting the majority class. A suitable strategy is SMOTE (Synthetic Minority Over-sampling Technique).

Justification

- SMOTE increases representation of the minority diabetic class by creating synthetic training examples.
- It can improve the model's ability to learn patterns related to diabetic patients.
- This is useful because missed diabetes cases are clinically important.

SMOTE should be applied only to the training data after the train-test split to avoid data leakage. Class weights are also a suitable alternative when synthetic oversampling is not preferred.

(c) 70:30 Split and Preprocessing

The dataset is divided into 70% training data and 30% testing data. The training portion provides enough examples for learning, while the larger 30% test set provides a meaningful independent evaluation for a medical use-case.

Preprocessing Steps

- Normalisation: Scale numerical variables such as age, glucose, BMI, blood pressure and cholesterol to comparable ranges when required by the selected model.
- Encoding: Convert categorical variables such as family history into numerical form using label encoding or one-hot encoding.
- Missing-value handling: Check for missing records and apply an appropriate imputation method when necessary.
- Stratified split: Preserve the diabetic/non-diabetic class distribution in both training and testing sets.

These steps improve data consistency and help prevent features with different scales or categories from being handled incorrectly.

TASK 2 – DECISION TREE FOR DIAGNOSIS

(a) Build a Decision Tree and First Three Levels

A Decision Tree classifier can be trained on the prepared 70% training dataset. The tree recursively selects the feature that gives the best split according to Information Gain or the Gini Index.

Illustrative First Three Levels

```
Level 1:            Glucose
                    /      \
Level 2:          BMI      Age
                  / \      / \
Level 3:      Outcome ... Outcome ...
              /   \
         Family History
          /          \
      Outcome      Outcome
```

The exact tree structure and root feature depend on the actual dataset and trained model. Therefore, the uploaded assessment specifies the method but does not provide enough data to calculate actual Information Gain or Gini values.

Root Node Justification

The root node is the feature with the largest Information Gain or greatest Gini impurity reduction among candidate features. For example, if Glucose gives the highest impurity reduction on the actual training data, it is selected as the root node.

(b) Model Evaluation and False Negatives

The Decision Tree is evaluated using Accuracy, Precision, Recall and F1-Score.

- Accuracy: Overall proportion of correct predictions.
- Precision: Of patients predicted as diabetic, how many are actually diabetic.
- Recall: Of actual diabetic patients, how many are correctly identified.
- F1-Score: Harmonic mean of Precision and Recall.

False Negatives are cases where the patient actually has diabetes but the model predicts non-diabetic. These are especially important clinically because the patient may miss early diagnosis and timely intervention.

Reducing False Negatives

- Use class weights or SMOTE to improve learning of the diabetic class.
- Tune the model and adjust the decision threshold where clinically appropriate.
- Use Recall and F1-score, not accuracy alone, when selecting the model.
- Validate the model on an independent dataset before deployment.

Clinically, reducing missed diabetes cases is important because early detection can support earlier monitoring and treatment.

(c) Post-Pruning and Clinical Deployment

Cost-complexity post-pruning removes branches from a fully grown tree when their additional complexity does not justify the improvement in performance.

Comparison

- Pre-pruning: Stops tree growth early using limits such as maximum depth or minimum samples per leaf. It is simpler but may underfit.
- Post-pruning: First grows the tree and then removes unnecessary branches. It can reduce overfitting while retaining useful splits.

Actual pre-pruning and post-pruning accuracy values cannot be reported without running both models on the provided dataset. For deployment, the preferred model should be selected using validation results, generalisation performance, recall for diabetic cases, and interpretability. In a clinical setting, a simpler well-validated pruned model can be preferable when it maintains safe diagnostic performance.

TASK 3 – STATISTICAL LEARNING FOR RISK STRATIFICATION

(a) Logistic Regression Comparison

Logistic Regression is selected as the statistical learning model. It estimates the probability that a patient belongs to the diabetic class using the patient features.

Comparison with Decision Tree

- Decision Tree: Can capture non-linear, rule-based relationships and is easy to interpret as decision rules.
- Logistic Regression: Provides probabilistic predictions and coefficients that show the direction and strength of feature effects.
- Accuracy: Compare the percentage of correct predictions on the same test set.
- AUC-ROC: Compare the ability of both models to distinguish diabetic and non-diabetic patients across thresholds.

The exact Accuracy and AUC-ROC values require the actual dataset and model execution. A statistical model may be preferable when calibrated probabilities, coefficient-based interpretation and a relatively simple global model are important.

(b) Feature Importance Analysis

Feature importance should be calculated from both trained models on the same dataset.

- Decision Tree: Use feature_importances_ to rank features according to their contribution to impurity reduction.
- Logistic Regression: Compare standardized absolute coefficient magnitudes, while also considering coefficient direction.

The top three predictors cannot be honestly identified from the assessment file alone because no patient dataset or trained output is provided. After execution, the top three features from both rankings should be listed and compared.

Agreement Justification

If both models rank the same features highly, this provides stronger evidence that those variables consistently contribute to prediction. If they disagree, possible reasons include non-linear relationships captured by the tree, feature correlation, scaling effects and different model assumptions.

TASK 4 – REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

(a) MDP Formulation

States

States represent patient health stages. For example: High Risk, Moderate Risk, Controlled and Improved. A state can be determined from relevant clinical measurements and diagnosis status.

Actions

The available actions are Diet, Exercise, Medication and Monitor.

Reward Function

A reward represents health improvement. Positive reward is given for movement toward a healthier state, while negative reward can be assigned for deterioration, unsafe outcomes or ineffective actions. The reward design should prioritise patient safety.

Transition Dynamics

Transition dynamics represent the probability of moving from one health state to another after an action. For example, a patient in a High Risk state may move to Moderate Risk, remain unchanged or worsen depending on the selected action and observed response.

Justification

This MDP structure directly represents the assessment scenario: patient condition is the state, treatment recommendation is the action, health improvement is the reward, and health changes over time are transitions.

(b) Q-Learning for 5 Patient Episodes

Q-Learning updates the action value using:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \, [\, r + \gamma \max Q(s',a') - Q(s,a) \,]$$

Below is an illustrative update process. Exact numerical values depend on the chosen transition and reward data.

Example Hyperparameters

- Learning rate $\alpha = 0.1$
- Discount factor $\gamma = 0.9$
- Exploration rate $\varepsilon = 0.2$

Illustrative Episodes

Episode 1: High Risk → Medication → Moderate Risk → Reward +4
Episode 2: Moderate Risk → Exercise → Controlled → Reward +5

Episode 3: High Risk → Diet → Moderate Risk → Reward +3
Episode 4: Controlled → Monitor → Controlled → Reward +2
Episode 5: Moderate Risk → Medication → Improved → Reward +6

Illustrative Q-table Updates Across Two Iterations

State-Action Pair	Iteration 1	Iteration 2
Q(High Risk, Medication)	0.00	0.40
Q(Moderate Risk, Exercise)	0.00	0.50
Q(Controlled, Monitor)	0.00	0.20

These values are an illustrative example of how Q-values can increase as beneficial actions receive positive rewards. Actual Q-values must be calculated from the actual state transitions and rewards used during implementation.

Convergence Criterion

Training can be considered converged when the maximum change in Q-values between successive iterations falls below a small threshold, such as 0.001, for a sustained number of episodes, and the learned policy remains stable.

(c) Final Policy and Ethical Considerations

Illustrative Final Learned Policy

High Risk → Medication
Moderate Risk → Exercise
Controlled → Monitor
Improved → Diet / Monitor

The actual final policy depends on the learned Q-table after training.

Ethical Considerations

- Patient Safety: The RL system must not independently make unsafe treatment decisions. Clinical oversight and safety constraints are required.
- Bias: Training data may contain demographic or clinical bias. The system should be tested for unfair performance across relevant patient groups.
- Explainability: Clinicians should understand why an action is recommended and be able to review the evidence.
- Human Oversight: Treatment recommendations should support qualified medical professionals rather than replace them.
- Validation: The system requires rigorous offline and clinical validation before any real-world deployment.
- Privacy and Governance: Patient data must be handled according to applicable privacy, security and healthcare requirements.

FINAL CONCLUSION

The proposed solution uses inductive learning and data preparation for diabetes prediction, a Decision Tree for interpretable diagnosis, Logistic Regression for statistical risk stratification, and Q-Learning for sequential treatment recommendation. Because the uploaded assessment does not include the actual patient dataset, exact trained metrics, tree values, feature rankings and learned Q-values must be generated by executing the models on the selected dataset. The structure above addresses all four tasks and clearly separates required methodology from results that require real data.